M. SUBHAN AHMAD

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BSCS student at FAST-NUCES focused on AI/ML, software development, and practical problem solving. Built semester projects in AI, DSA, OOP, probability/statistics, and operating systems.

Education

Bachelor of Science in Computer Science (BSCS) | National University of Computer and Emerging Sciences (FAST-NUCES) | 4th Semester | Expected 2028

Relevant Coursework: Programming Fundamentals, Object-Oriented Programming, Data Structures, Discrete Mathematics, Linear Algebra, Probability & Statistics, Introduction to Machine Learning.

Technical Skills

Languages: Python, C++, SQL, HTML, CSS

Frameworks / Libraries: Flask, Streamlit, Pandas, NumPy, scikit-learn

Concepts: OOP, DSA, Graph Algorithms, Search Algorithms, CSP, Machine Learning, Probability & Statistics

Tools: Git/GitHub, VS Code, Jupyter

Projects

AI Semester Project

Python | Tkinter | NumPy | scikit-learn | Matplotlib

- Built a four-module AI semester project covering intelligent urban delivery robot simulation, Sudoku CSP solving, automated exam management, and learning-in-AI experiments on the Iris dataset.
- Compared BFS, DFS, UCS, Greedy Best-First, and A* search; implemented AC3 plus backtracking; trained perceptron and delta-rule models with experiment reporting.

Concurrent Banking System - OS Final Project

C++ | Multithreading | Synchronization | Bankers Algorithm | IPC | Memory Management

GitHub: github.com/Subhan3716/concurrent-banking-os-project

- Designed a menu-driven operating systems project with CPU scheduling (FCFS, Priority, Round Robin), synchronization, deadlock avoidance, IPC, and memory management modules.
- Generated scheduling Gantt charts, synchronization logs, IPC flow outputs, and FIFO/LRU memory comparison metrics for the final report.

CampusLink: Integrated University Management System

C++ | Data Structures and Algorithms

- Built a university management system using hash tables, graphs, AVL trees, queues, and stacks.
- Implemented Dijkstra and Prim's algorithms and organized the codebase into modular components for clearer separation of concerns.

Kingdom Management Simulator OOP

C++ | Object-Oriented Programming

- Designed a console-based OOP simulator for managing population, economy, army, banking, politics, resources, and trade.
- Structured the project with classes and modules to keep gameplay logic maintainable and easy to extend.

Pakistan Bus Delay Predictor

Python | Streamlit | Pandas | NumPy | scikit-learn

- Built a Streamlit app to predict intercity bus delays in Pakistan using probability, statistics, feature engineering, and machine learning.
- Evaluated multiple models and surfaced the best-performing prediction workflow through an interactive web interface.

Experience

Freelance Software Developer | Jan 2024 - Present

- Delivered small web and automation projects using Python, HTML, CSS, and JavaScript-style workflows.
- Worked on client requirements, debugging, and quick iterations while keeping communication clear and deadlines on track.

Extracurricular

- Active in university coding communities, hackathons, and competitive problem-solving activities.
- Interested in AI/ML, software engineering, and building practical systems that solve real-world problems.